

Problems on Numbers

1. A number is as much greater than 36 as is less than 86. Find the number.

A) 51

B) 61

C) 71

D) 81

Answer: B) 61

2. Find a number such that when 15 is subtracted from 7 times the number, the result is 10 more than twice the number.

A) 5

B) 15

C) 7.5

D) 4

Answer: A) 5

3. The sum of two numbers is 25 and their difference is 13. Find their product.

A) 104

B) 114

C) 315

D) 325

Answer: B) 114

4. If one-third of one-fourth of a number is 15, then three-tenth of that number is:

A) 35

B) 36

C) 45

D) 54

Answer: D) 54

5. The sum of two numbers is 40 and their product is 375. What is the sum of their reciprocals?

A) $\frac{8}{75}$

B) $\frac{1}{40}$

C) $\frac{75}{8}$

D) $\frac{75}{4}$

Answer: A) $\frac{8}{75}$

6. The difference between a two-digit number and the number obtained by interchanging the positions of its digits is 36. What is the difference between the two digits of that number?

A) 3

B) 4

C) 9

D) Cannot be determined

Answer: B) 4

7. The sum of the squares of three consecutive odd numbers is 2531. Find the numbers.

A) 25, 27, 29

B) 27, 29, 31

C) 29, 31, 33

D) 23, 25, 27

Answer: B) 27, 29, 31

8. Two numbers are in the ratio 3:5. If 9 is subtracted from each, the new numbers are in the ratio 12:23. The smaller number is:

A) 27

B) 33

C) 49

D) 55

Answer: B) 33

9. The product of two numbers is 120 and the sum of their squares is 289. The sum of the numbers is:

- A) 20
- B) 23
- C) 169
- D) 29

Answer: B) 23

10. A number consists of two digits. If the digits interchange places and the new number is added to the original number, the resulting sum will be divisible by:

- A) 3
- B) 5
- C) 9
- D) 11

Answer: D) 11

11. If 20% of a number is added to 20% of 150, the result is 60. Find the number.

- A) 100
- B) 120
- C) 150
- D) 180

Answer: C) 150

12. The sum of the digits of a two-digit number is 15 and the difference between the digits is 3. What is the two-digit number?

- A) 69
- B) 78
- C) 96
- D) Cannot be determined

Answer: D) Cannot be determined (Could be 96 or 69)

13. Three times the first of three consecutive odd integers is 3 more than twice the third. The third integer is:

- A) 9

B) 11

C) 13

D) 15

Answer: D) 15

14. The difference between a positive proper fraction and its reciprocal is $\frac{9}{20}$. The fraction is:

A) $\frac{3}{5}$

B) $\frac{3}{10}$

C) $\frac{4}{5}$

D) $\frac{5}{4}$

Answer: C) $\frac{4}{5}$

15. A number when multiplied by 13 is increased by 180. The number is:

A) 5

B) 12

C) 15

D) 45

Answer: C) 15

16. The sum of three consecutive even numbers is 24 more than the first of these numbers. The middle number is:

A) 12

B) 14

C) 16

D) 18

Answer: B) 14

17. A number is 25 more than its $\frac{2}{5}$ th. The number is:

A) 60

B) 80

C) $\frac{125}{3}$

D) $125/7$

Answer: C) $125/3$

18. If the sum of two numbers is 42 and their product is 437, then the absolute difference between the numbers is:

A) 3

B) 4

C) 5

D) 7

Answer: B) 4

19. The ratio between a two-digit number and the sum of the digits of that number is 4:1. If the digit in the unit's place is 3 more than the digit in the ten's place, then the number is:

A) 24

B) 36

C) 48

D) 63

Answer: B) 36

20. Which number is 40% less than 90?

A) 36

B) 54

C) 50

D) 60

Answer: B) 54

21. The sum of two numbers is 80. If one number is $2/3$ of the other, find the smaller number.

A) 24

B) 32

C) 40

D) 48

Answer: B) 32

22. A fourth of a number exceeds a seventh of itself by 24. The number is:

A) 224

B) 254

C) 284

D) 324

Answer: A) 224

23. 24 is divided into two parts such that 7 times the first part added to 5 times the second part makes 146. The first part is:

A) 11

B) 13

C) 16

D) 17

Answer: B) 13

24. The sum of a rational number and its reciprocal is $\frac{13}{6}$. Find the number.

A) $\frac{2}{3}$

B) $\frac{3}{2}$

C) $\frac{4}{3}$

D) $\frac{5}{2}$

Answer: B) $\frac{3}{2}$

25. The product of two consecutive positive integers is 240. The numbers are:

A) 14, 15

B) 15, 16

C) 16, 17

D) 18, 19

Answer: B) 15, 16

26. 50 is divided into two parts such that the sum of their reciprocals is $\frac{1}{12}$. The two parts are:

A) 20, 30

B) 24, 26

C) 28, 22

D) 36, 14

Answer: A) 20, 30

27. The difference of two numbers is 11 and one-fifth of their sum is 9. The numbers are:

A) 31, 20

B) 30, 19

C) 29, 18

D) 28, 17

Answer: D) 28, 17

28. The sum of the squares of two numbers is 3341 and the difference of their squares is 891. The numbers are:

A) 25, 46

B) 35, 46

C) 25, 36

D) 35, 36

Answer: B) 35, 46

29. Three numbers are in the ratio 4:5:6 and their average is 25. The largest number is:

A) 30

B) 32

C) 36

D) 42

Answer: A) 30

30. A number exceeds 20% of itself by 40. The number is:

A) 50

B) 60

C) 80

D) 48

Answer: A) 50

31. If $\frac{1}{5}$ th of a number is 20 less than $\frac{1}{3}$ rd of the same number, then the number is:

A) 50

B) 75

C) 150

D) 200

Answer: C) 150

32. The difference between a number and its $\frac{3}{5}$ is 50. What is the number?

A) 75

B) 100

C) 125

D) 150

Answer: C) 125

33. If the sum of two numbers is 10 and the sum of their reciprocals is $\frac{5}{12}$, the numbers are:

A) 8, 2

B) 6, 4

C) 7, 3

D) 9, 1

Answer: B) 6, 4

34. A number multiplied by 5 exceeds 20 by the same amount as it exceeds 40 when multiplied by 3. The number is:

A) 5

B) 10

C) 15

D) 20

Answer: C) 15

35. The sum of 5 consecutive numbers is 190. What is the sum of the largest and smallest numbers?

A) 75

B) 76

C) 77

D) 78

Answer: B) 76

36. Which whole number is such that when half of it is added to itself, the sum is 15?

A) 8

B) 10

C) 12

D) 14

Answer: B) 10

37. The denominator of a fraction exceeds the numerator by 2. If 1 is added to both, the ratio becomes 2:3. The fraction is:

A) $\frac{3}{5}$

B) $\frac{5}{7}$

C) $\frac{7}{9}$

D) $\frac{1}{3}$

Answer: A) $\frac{3}{5}$

38. The sum of the squares of three numbers is 138, while the sum of their products taken two at a time is 131. Their sum is:

A) 20

B) 30

C) 40

D) 50

Answer: A) 20

39. A number consists of two digits. The sum of the digits is 9. If 63 is subtracted from the number, its digits are interchanged. Find the number.

A) 63

B) 72

C) 81

D) 90

Answer: C) 81

40. If the sum of a number and its square is 182, what is the number?

A) 12

B) 13

C) 14

D) 15

Answer: B) 13

41. $\frac{4}{5}$ of a certain number is 64. Half of that number is:

A) 32

B) 40

C) 80

D) 16

Answer: B) 40

42. What is the number which when multiplied by 15 is increased by 196?

A) 14

B) 20

C) 26

D) 28

Answer: A) 14

43. The difference between the squares of two consecutive odd integers is always divisible by:

A) 3

B) 6

C) 7

D) 8

Answer: D) 8

44. The sum of the digits of a 2-digit number is 10. If the number formed by reversing the digits is less than the original number by 36, find the number.

A) 73

B) 64

C) 82

D) 91

Answer: A) 73

45. Two numbers are such that the square of one is 224 less than 8 times the square of the other. If the numbers are in the ratio 3:4, they are:

A) 6, 8

B) 9, 12

C) 12, 16

D) 15, 20

Answer: A) 6, 8

46. Find the number which when added to itself 13 times, gives 112.

A) 7

B) 8

C) 9

D) 11

Answer: B) 8

47. Two numbers are in the ratio 5:7. If 9 is subtracted from each, they are in the ratio 7:11. The difference of the numbers is:

A) 6

B) 12

C) 15

D) 18

Answer: B) 12

48. If the numerator of a fraction is increased by 200% and the denominator is increased by 350%, the resultant fraction is $\frac{5}{12}$. What was the original fraction?

A) $\frac{5}{8}$

B) $\frac{5}{9}$

C) $\frac{7}{12}$

D) $\frac{11}{12}$

Answer: A) $\frac{5}{8}$

49. Three numbers are such that the second is twice the first and is also thrice the third. If the average of the three numbers is 44, the largest number is:

A) 24

B) 36

C) 72

D) 108

Answer: C) 72

50. The sum of three consecutive odd numbers is 20 more than the first of these numbers. What is the middle number?

A) 7

B) 9

C) 11

D) 13

Answer: B) 9

51. A number is 4 less than 3 times another number. If their sum is 16, find the numbers.

A) 3, 13

B) 4, 12

C) 5, 11

D) 6, 10

Answer: C) 5, 11

52. The product of two numbers is 192 and the sum of these two numbers is 28. What is the smaller of these two numbers?

A) 12

B) 14

C) 16

D) 18

Answer: A) 12

53. The sum of the squares of two consecutive even integers is 340. The numbers are:

A) 10, 12

B) 12, 14

C) 14, 16

D) 16, 18

Answer: B) 12, 14

54. If $\frac{3}{4}$ of a number is 7 more than $\frac{1}{6}$ of the number, then $\frac{5}{3}$ of the number is:

A) 12

B) 18

C) 15

D) 20

Answer: D) 20

55. Dividing a number by 56, we get 29 as remainder. On dividing the same number by 8, what will be the remainder?

A) 4

B) 5

C) 6

D) 7

Answer: B) 5

56. The difference of two numbers is 20% of the larger number. If the smaller number is 20, the larger number is:

A) 25

B) 45

C) 50

D) 80

Answer: A) 25

57. Which number is such that its $\frac{1}{4}$ th added to its $\frac{1}{5}$ th makes 9?

A) 15

B) 20

C) 25

D) 30

Answer: B) 20

58. Two-third of a number is 20 more than 40% of the same number. The number is:

A) 50

B) 60

C) 75

D) 80

Answer: C) 75

59. $\frac{5}{8}$ of a number is 20 more than $\frac{3}{8}$ of the number. Find the number.

A) 60

B) 80

C) 100

D) 120

Answer: B) 80

60. The sum of 7 consecutive numbers is 175. The largest number is:

A) 26

B) 28

C) 24

D) 30

Answer: B) 28

61. 96 is divided into two parts in such a way that the seventh part of the first and ninth part of the second are equal. The smallest part is:

A) 38

B) 40

C) 42

D) 48

Answer: C) 42

62. A number when divided by 899 gives a remainder 63. If the same number is divided by 29, the remainder will be:

A) 5

B) 10

C) 23

D) 25

Answer: A) 5

63. The sum of three consecutive multiples of 3 is 72. The largest number is:

A) 21

B) 24

C) 27

D) 30

Answer: C) 27

64. The sum of digits of a two-digit number is 12. The digits differ by 2. The number is:

A) 57

B) 75

C) Both A and B

D) None of these

Answer: C) Both A and B

65. A number consists of two digits whose sum is 8. If 18 is added to the number, its digits are reversed. The number is:

A) 35

B) 53

C) 26

D) 62

Answer: A) 35

66. A fraction becomes 4 when 1 is added to both the numerator and denominator and it becomes 7 when 1 is subtracted from both the numerator and denominator. The numerator of the given fraction is:

A) 2

B) 3

C) 7

D) 15

Answer: D) 15

67. If the sum of two numbers is 33 and their difference is 15, the smaller number is:

A) 9

B) 12

C) 15

D) 18

Answer: A) 9

68. The sum of a number and its reciprocal is 2. The number is:

A) 0

B) 1

C) -1

D) 2

Answer: B) 1

69. The product of two numbers is 3200 and the quotient when the larger number is divided by the smaller is 2. The numbers are:

A) 16, 32

B) 32, 64

C) 40, 80

D) 20, 40

Answer: C) 40, 80

70. The sum of two numbers is 22. Five times one number is equal to 6 times the other. The bigger of the two numbers is:

A) 10

B) 12

C) 15

D) 16

Answer: B) 12

71. A number is such that the ratio of $\frac{3}{5}$ th of the number to the $\frac{2}{7}$ th of the number is 21:10. The number is:

A) 50

B) 75

C) 100

D) Cannot be determined

Answer: D) Cannot be determined (Any number satisfies this ratio)

72. One-fourth of the sum of two numbers is 18 and half of their difference is 11. The numbers are:

A) 47, 25

B) 37, 23

C) 42, 28

D) 50, 22

Answer: A) 47, 25

73. The ratio of two numbers is 3:4 and their sum is 420. The greater of the two numbers is:

A) 180

B) 240

C) 360

D) 400

Answer: B) 240

74. If 50 is subtracted from two-third of a number, the result is equal to sum of 40 and one-fourth of that number. What is the number?

A) 174

B) 216

C) 246

D) 336

Answer: B) 216

75. The sum of the squares of two numbers is 80 and the square of their difference is 36. The product of the two numbers is:

A) 22

B) 44

C) 58

D) 116

Answer: A) 22

76. A number whose fifth part increased by 4 is equal to its fourth part diminished by 10, is:

A) 240

B) 260

C) 270

D) 280

Answer: D) 280

77. Three numbers are in the ratio 1:2:3 and their H.C.F is 12. The numbers are:

- A) 4, 8, 12
- B) 5, 10, 15
- C) 10, 20, 30
- D) 12, 24, 36

Answer: D) 12, 24, 36

78. The sum of three numbers is 98. If the ratio of the first to the second is 2:3 and that of the second to the third is 5:8, then the second number is:

- A) 20
- B) 30
- C) 48
- D) 58

Answer: B) 30

79. The sum of two numbers is 184. If one-third of the one exceeds one-seventh of the other by 8, find the smaller number.

- A) 64
- B) 72
- C) 76
- D) 84

Answer: B) 72

80. Two numbers are such that their difference, their sum and their product are in the ratio of 1:7:24. The product of the numbers is:

- A) 6
- B) 12
- C) 24
- D) 48

Answer: D) 48

81. A number is 30 less than its square. The number is:

A) 4

B) 5

C) 6

D) 7

Answer: C) 6

82. In a division sum, the divisor is 10 times the quotient and 5 times the remainder. If the remainder is 46, what is the dividend?

A) 5336

B) 5363

C) 5636

D) 5346

Answer: A) 5336

83. If the sum of two numbers is 14 and their difference is 10, find the product of these two numbers.

A) 18

B) 20

C) 24

D) 22

Answer: C) 24

84. Find the largest number of 4 digits divisible by 12, 15, 18 and 27.

A) 9690

B) 9720

C) 9930

D) 9960

Answer: B) 9720

85. A number is increased by 10% and then decreased by 10%. The net change is:

A) 0%

B) 1% increase

C) 1% decrease

D) 2% decrease

Answer: C) 1% decrease

86. The sum of seven consecutive numbers is 161. The largest of these numbers is:

A) 21

B) 23

C) 25

D) 26

Answer: D) 26

87. If the product of two numbers is 324 and their H.C.F is 3, then their L.C.M is:

A) 98

B) 108

C) 112

D) 118

Answer: B) 108

88. 40% of a number is equal to $\frac{3}{5}$ th of another number. What is the ratio between the first number and the second number?

A) 3:2

B) 2:3

C) 3:4

D) 4:3

Answer: A) 3:2

89. A number is 56 more than the average of its half, its third and its sixth. The number is:

A) 84

B) 112

C) 168

D) 336

Answer: C) 168

90. The sum of two numbers is 25 and their difference is 13. Find their product.

A) 104

B) 114

C) 315

D) 325

Answer: B) 114

91. If a number is subtracted from its square, the remainder is 20. The number is:

A) 3

B) 4

C) 5

D) 6

Answer: C) 5

92. $\frac{5}{6}$ of a number is 60 more than $\frac{1}{2}$ of the number. The number is:

A) 120

B) 150

C) 180

D) 210

Answer: C) 180

93. The sum of two numbers is 37 and the difference of their squares is 185. The difference between the two numbers is:

A) 3

B) 4

C) 5

D) 7

Answer: C) 5

94. What is the sum of two consecutive even numbers, the difference of whose squares is 84?

A) 34

B) 38

C) 42

D) 46

Answer: C) 42

95. If the sum of two numbers is 50 and their mean proportional is 24, then the numbers are:

A) 36, 14

B) 32, 18

C) 40, 10

D) 42, 8

Answer: B) 32, 18

96. A two-digit number is 4 times the sum of its digits. If 18 is added to the number, the digits are reversed. The number is:

A) 12

B) 24

C) 36

D) 48

Answer: B) 24

97. If a number is doubled and then 20 is added to it, the result is 3 times the number. The number is:

A) 10

B) 20

C) 30

D) 40

Answer: B) 20

98. The sum of a number and its square is 72. The number is:

A) 6

B) 7

C) 8

D) 9

Answer: C) 8

99. Two numbers differ by 5. If their product is 336, the sum of the two numbers is:

A) 21

B) 28

C) 37

D) 51

Answer: C) 37

100. A number when divided by 6 leaves a remainder 3. When the square of the same number is divided by 6, the remainder is:

A) 0

B) 1

C) 2

D) 3

Answer: D) 3